

Sarah Agrawal

858-205-7412 | sarah.agrawal@berkeley.edu | [Portfolio](#) | linkedin.com/in/sarah-agrawal | github.com/sarah-agrawal

EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science and Applied Mathematics

Expected May 2029

- **Coursework:** Data Structures & Algorithms, Artificial Intelligence, Structure & Interpretation of Computer Programs, Internet Architecture & Protocols, Principles of Data Science, Full Stack Web Development, Discrete Math, Linear Algebra
- **Organizations:** Women in Computing & Data Science, Math Undergraduate Student Association, Claude Builder Club

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, TypeScript, SQL, HTML/CSS, Scheme

AI/ML: PyTorch, XGBoost, scikit-learn, SHAP, NumPy, Pandas, Matplotlib

Frameworks: React.js, Next.js, Redux, Node.js, Express.js, Flask, Django, Tailwind CSS, Three.js, Streamlit

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Supabase, SQLAlchemy

Developer Tools: Git, GitHub, Linux/Unix, npm, REST APIs, Vercel, Jupyter, Jetty, JUnit, Figma, Claude Code

EXPERIENCE

Data Science & AI Researcher

May 2026 – Present

Filmcrowd

Berkeley, CA

- Building FilmCrowd's **Greenlight Score**, an **ML model** that rates a script's commercial potential for investors
- Laid the data foundation for the score by compiling **1M+ films** from TMDb and enriching them with Box Office Mojo, IMDb and Rotten Tomatoes data on budget, revenue, cast and critic scores
- Ran **regression, correlation and ANOVA tests** in **Python** on **29,000+ films** to find which factors drive box-office revenue, selecting the features the score is built on
- Interviewed industry veterans including producer **Barrie Osborne** (*The Lord of the Rings*) and Weta Digital VFX supervisor **Jon Labrie** to capture how films get greenlit, guiding how the score weighs each factor

Machine Learning Researcher

Dec 2023 – Present

Carnegie Mellon University, with Dr. Ganesh Mani

Remote

- Built **RallyIQ** (live app, [code](#)), a shot-outcome model trained on **81,000+ strokes** from **94 pro badminton matches**
- Trained a **causal Transformer** in **PyTorch** over rally sequences alongside **XGBoost**; the ensemble reached **0.84 AUC** on winning shots across **19 held-out matches**, cutting log loss **13%** below a base-rate model
- Shipped a **Streamlit** app that replays rallies and recommends the highest-value realistic shot, with **SHAP** explanations

PROJECTS

Build Your Own World | *Java, procedural generation, StdDraw*

Jul. 2026

- Built a **procedural world generator** that carves connected rooms and hallways from a seeded random number generator, guaranteeing every floor tile is reachable and every seed reproduces the same world
- Built a **real-time tile renderer** with keyboard movement, a HUD, **save/load serialization** and procedural music

Ngordnet | *Java, Jetty, graphs, TreeMap*

Jun. 2026

- Designed a **TreeMap**-backed time-series type and an **NGramMap** index that parses a **300MB Google Books corpus** in a single pass, so repeated word-frequency queries never re-read the data
- Built a **directed graph** and **WordNet parser** from scratch, using graph traversal to resolve hyponyms across multi-word queries with frequency-ranked top-k filtering, served through a **Jetty REST API** to a JavaScript frontend

Scheme Interpreter | *Python, recursion, interpreters*

Apr. 2025

- Implemented a **Scheme interpreter** in **Python** with a mutually recursive **eval/apply** loop, **environment frames** for lexical scoping, and **8 special forms** including **define**, **lambda** and **cond**

LEADERSHIP

Founder, ConnectUp (2024 – Present): founded a mentorship initiative for students moving to the US, guiding **25+** students through the academic, social and cultural shift in one-on-one video calls

President, CCA Key Club (2022 – 2025): grew membership to **100+**, led **400+** logged service hours, ran food drives collecting **3,000+ lbs** and raised **\$1,000+** with Del Mar Kiwanis

Director of Events, CCA Women in STEM (2022 – 2025): created math lessons in Dari for **20+** girls in Afghanistan barred from school, organized **10** fundraisers and STEM events, and grew membership by **25+**

Museum Ambassador, Fleet Science Center (2022 – Present): led **5** STEM camps for **125+** kids building robots from e-waste, ran **20+** live science demos and volunteered **450+** hours